

Title

Reduction in heavy drinking episodes but not total consumption with direct current neuromodulation

Abstract

Objective: To assess the efficacy of direct current brain stimulation versus sham control in reducing alcohol consumption in adults with established alcohol use disorder. **Setting:** 14 outpatient clinical sites in France and Monaco from 2015-2022. **Participants:** 337 randomized adults with alcohol use disorder (predominantly male, mean age 51 years). **Intervention:** 2-mA direct current applied to prefrontal regions daily for 5 days versus sham. **Results:** At 24 weeks, the active stimulation group showed a statistically significant decrease in days per month with heavy drinking episodes (mean reduction 2.5 days per 4-week period, 97.5% CI: -4.9 to -0.1, $p=0.022$) compared to sham. However, total daily alcohol volume reduction was not statistically significant between groups (-6 g/day, 97.5% CI: -15.2 to 3.3, $p=0.15$). The intervention was tolerated well, with similar adverse event rates between groups. **Conclusions:** Active stimulation demonstrated modest, sustained benefit for reducing drinking frequency but not absolute consumption volume.